

WEEK 6 — The “Story”: You are designing a system, and the user is your “material”

Imagine you’re building an app/system. Week 6 is basically saying:

1. Design is **not just making screens**. It’s designing the *whole interaction* + preventing mistakes.
 2. You must **know your user** (they are NOT you).
 3. You must plan **navigation** so users never feel lost.
 4. You must design screens using layout rules (grouping, order, alignment, whitespace).
 5. And all of this works only if you understand **human memory and thinking** (STM limits, errors, emotion).
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1) Design interactions + interventions (not just interfaces)

1.1 “Interactions” not just “interfaces”

The slide gives a stapler example:

- Manual workflow: write, print, staple, write, print, staple...
- Electronic workflow: write, print, write, print... staple at the end.

Meaning: technology changes the *pattern* of work. It’s not only about the screen, it changes how people behave.

1.2 “Interventions” not just “artefacts”

Design isn’t only the system itself, it includes:

- manuals, tutorials, documentation
- what you *tell* users and how you support them

Example: A complicated app with an excellent onboarding tutorial can feel usable. Same app without help feels impossible.

👉 **MEMORIZE:**

- Design interactions, not just interfaces.
- Design interventions, not just artefacts.

2) What is design? “Goals within constraints”

2.1 Definition

Design = achieving **goals** within **constraints**.

- **Goals:** who is it for? why do they want it?
- **Constraints:** platform, materials, time, budget, technical limits
- **Trade-offs:** you can't maximize everything at once.

👉 **MEMORIZE** (perfect exam line):

Design = achieving goals within constraints (through trade-offs).

3) Golden rule of design: “Understand your materials”

This is one of the most testable lines.

3.1 Golden rule

Understand your materials.

For HCI, your materials are:

1. **Computers:** limitations, capacities, tools, platforms
2. **People:** psychology, social behavior, human error
3. **Their interaction:** how people and tech clash/fit together

Story way: If you don't understand wood, you can't build a chair. If you don't understand people + computers, you can't build a usable system.

👉 **MEMORIZE:** “Understand your materials” + list computers/people/interaction.

4) “To err is human” = errors are design problems, not user problems

This section is your teacher's favorite because it's a philosophy.

Accident reports often blame “human error.”

But the lecture argues: if a concrete lintel breaks, we blame **design**, not “lintel error.” Similarly, when humans slip, it’s often **design error**.

Key idea:

- Human error is **normal and predictable**, especially under stress.
- Since we know users make errors, we should **design for it**.

Example (modes + lock story):

A phone has a “lock” feature. If lock is on and you press “yes”, it might open phone book; inside phone book “c” means delete entry and “yes” confirms → oops, accidental deletion.

This shows:

- Context matters (same button = different meaning in different mode).
- Modes can cause dangerous slips.

 MEMORIZE:

- “Human error is normal → design for it.”
 - Modes can create accidental actions.
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5) The design process: not linear — iterative

5.1 Core message

You never get it right first time.

Process steps (the lecture’s version):

1. Requirements (what’s wanted, what exists)
2. Analysis (understand and organize)
3. Design (decide what to do + how)
4. Iteration & prototyping (test, fix, discover what’s really needed)
5. Implementation & deployment (build and release)

Important nuance:

Time is limited, so you make **trade-offs**: too much design effort can waste time, too little design gives bad usability.

5.2 Pitfall of prototyping (“hill climbing” problem)

If you keep improving a bad initial idea, you might reach a “small hill” (local maximum) instead of the best solution.

So you need:

1. a good starting point
2. understand what’s wrong so you can fix the *right* things

👉 MEMORIZE:

- Iteration: never right first time
 - Prototyping pitfall: local maxima (Malverns vs Matterhorn metaphor)
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6) User focus: know your user (they’re not you)

6.1 Know your user

- Who are they? skills? goals? context?
- They’re **probably not like you**.
How to know them:
 - talk to them, watch them, use empathy/imagination

6.2 Personas (MEMORIZE definition)

Persona = a detailed description of an example user (not necessarily real), used as a surrogate.

Example persona in slides: Abdul Shakoor — background, skills, fear of new systems, physical limitation, etc.

Why details matter:

- It makes the persona feel real
- Helps designers ask: “What would Abdul do?” (instead of guessing)

6.3 Cultural probes

When observation is hard (private homes, sensitive patients), use “probe packs”:

- camera, postcards, objects that prompt recording in their natural setting

Goal:

- Not perfect objective data

- More like inspiration + deeper understanding of the user's world

👉 **MEMORIZE:**

- Persona definition
 - Cultural probes = probe packs to prompt participant self-recording in real environment
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7) Scenarios: stories for design (super testable)

Scenarios are **stories** describing how users will use the system.

Why use scenarios?

- communicate ideas to team/users
- validate other models
- understand step-by-step dynamics

How to write them (teacher nitpick):

- What do users want to do?
- Step-by-step walkthrough
- What they see (sketch/screenshots)
- What they do (click/type)
- What are they thinking?

Example scenario: video lectures:

- Student wants bookmarks/tags, share annotations
- Deaf student needs transcripts/captions; social web-based discussion helps keep up

From scenarios, extract requirements (the slide lists them): search topics, captions, bookmarks, linking, web links, filter annotations, feedback evaluation, etc.

Linearity pros/cons (MEMORIZE)

Scenarios are linear:

- Pros: natural, easy, concrete
 - Cons: don't show choices/branches/error conditions
Solution: use multiple scenarios + multiple methods
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8) Navigation design: local vs global vs wider

Navigation = helping users find their way.

Levels:

- **Local:** within a screen / from one screen out
- **Global:** across whole site/app structure
- **Wider:** between apps + environment (OS, web links)

8.1 Four golden rules of navigation (MEMORIZE)

At any point user should know:

1. Where you are
2. What you can do
3. Where you are going / what will happen
4. Where you've been / what you've done

Examples:

- "Where you are" → breadcrumbs

8.2 Structures: hierarchy vs network diagrams

- Hierarchical diagrams: functional separation into groups/screens
- Network diagrams: show branches and different paths; more task oriented

Deep hierarchy problem: "deep is difficult" (people misuse Miller's 7 ± 2 as menu-size rule).
Better to avoid deep drilling; structure within screens.

8.3 Modern navigation infrastructures (from your extra deck)

These are named patterns, likely MCQ material:

- **Nested Doll:** linear drilling down, good for small screens; bad for lateral movement
- **Hub and Spoke:** central hub; must return to hub to switch sections
- **Bento Box:** dashboard pulling multiple components on one screen
- **Filtered View:** one dataset explored via filters/sorts/views
- Hybrids

8.4 New navigation modalities (also likely MCQ)

- Touch/gesture (pinch, swipe, edge gestures, air/3D)
- Voice (assistant, conversational)

- Sensor/motion (tilt, shake)
 - Eye/gaze (dwell select)
 - Haptics (vibration cues)
 - Spatial/AR/VR navigation (teleportation, hand tracking)
 - BCI (EEG)
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9) Screen design & layout: make screens readable and scannable

This part is super “practical rules” and exam-friendly.

Core idea

Ask: what is the user doing? Form follows function.

Tools for layout:

- grouping, order, decoration, alignment, white space

9.1 Grouping & structure

Rule: **logically together** ⇒ **physically together**

Use:

- boxes, spacing, visual boundaries

9.2 Order

- Should match natural task order
- Set correct tab order
- Avoid “cake recipe syndrome” (steps shown in weird order that confuses)

9.3 Decoration

- Boxes to group
- Fonts for emphasis/heading
- But not too many fonts/styles

9.4 Alignment

- Text: left-align for left-to-right languages; centered looks fancy but harder to scan

- Names: align surnames if that's what people scan for
- Numbers: align decimal points or right-align to compare sizes

9.5 Multiple columns & white space

- Scanning across gaps is hard; use leaders or shading carefully
- White space is not empty; it separates, structures, highlights

9.6 Forms (entering information)

- Same layout rules apply
- Group logically
- Natural order top-bottom / left-right (culture dependent)
- Set tab order

9.7 Presenting information: aesthetics vs utility

- Aesthetically pleasing designs increase satisfaction and productivity, but beauty can conflict with readability
- Color and 3D are often abused:
 - beware color blindness
 - use color sparingly to reinforce info
 - avoid useless 3D (3D pie charts, perspective text)

9.8 Localization & internationalization (MEMORIZE)

Design across countries:

- Just translating text is not enough
- Layout sizes, direction, cultural meanings differ (tick/cross meanings vary)

10) Human cognition: memory, thinking, errors, emotion

This is the “why” behind UI rules.

10.1 Three memory systems (MEMORIZE)

1. Sensory memory
 - very brief buffers: iconic (visual), echoic (sound), haptic (touch)
 - overwritten continuously
2. Short-term / Working memory (STM)
 - “scratchpad”

- rapid access, rapid decay
- limited capacity: **7 ± 2 chunks**
- 3. Long-term memory (LTM)
 - huge repository
 - slow access but massive capacity
 - episodic (events) and semantic (facts/skills)

👉 **MEMORIZE:** Sensory, STM (7±2 chunks), LTM (episodic vs semantic).

10.2 Models of LTM

- Semantic networks (concepts linked)
- Frames (slots, defaults)
- Scripts (stereotyped situation info)
- Production rules (IF condition THEN action)

10.3 Forgetting

- decay (slow)
- interference:
 - retroactive (new replaces old)
 - proactive (old interferes with new)

10.4 Thinking & reasoning (MEMORIZE 3 types)

- Deduction: logically necessary from premises (can be logically valid but factually wrong if premise false)
- Induction: generalize seen → unseen (useful but unreliable; can be proven false not true)
- Abduction: infer cause from event (best explanation; can be wrong)

10.5 Problem solving

- Problem space theory: states + operators + heuristics
- Analogy: mapping known problem to new domain
- Skill acquisition: experts chunk information better; conceptual grouping

10.6 Errors: slips vs mistakes (MEMORIZE)

- **Slips:** right intention, wrong action (inattention, skill)
- **Mistakes:** wrong intention due to incorrect understanding (wrong mental model)

10.7 Emotion & affect (important link to design)

Affect influences performance:

- Positive affect → more creativity
- Negative affect (stress) → narrow thinking, harder tasks

Design implications:

- Stress increases difficulty of problem solving
- Relaxed users forgive design flaws
- Aesthetically pleasing interfaces increase positive affect

10.8 Individual differences (MEMORIZE idea)

Differences can be:

- long-term (abilities)
 - short-term (fatigue, stress)
 - changing (age)
- Design question: “Will this exclude some users?”
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WHAT TO MEMORIZE (Quick list)

MEMORIZE DEFINITIONS/LISTS:

- Design = goals within constraints (trade-offs)
 - Golden rule: understand your materials (computers, people, interaction)
 - “To err is human” → design for human error
 - Iterative process steps + “never right first time”
 - Persona definition + why details matter
 - Cultural probes definition
 - Scenario: what it is + what to include + linearity pros/cons
 - Four golden rules of navigation
 - Navigation infrastructures: nested doll, hub/spoke, bento, filtered view
 - Layout tools: grouping/order/decoration/alignment/white space
 - Memory model: sensory, STM (7 ± 2 chunks), LTM (episodic/semantic)
 - Reasoning: deduction, induction, abduction
 - Errors: slips vs mistakes
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PRACTICE QUESTIONS (Week 6)

A) 5 MCQs (with answers + explanation)

MCQ 1

Design is best defined as:

- A) Making screens look pretty
- B) Coding the final product
- C) Achieving goals within constraints
- D) Reducing all errors to zero

Answer: C

Explanation: The lecture defines design as achieving **goals within constraints**, which requires trade-offs.

MCQ 2

The “golden rule of design” in HCI is:

- A) Use bright colors
- B) Keep menus at 7 ± 2 items
- C) Understand your materials
- D) Always use prototypes

Answer: C

Explanation: The golden rule is “understand your materials,” and in HCI the materials are computers + people + their interaction.

MCQ 3

Which is a **slip**?

- A) User thinks “Save” means “Share,” then presses Share
- B) User intends to press “Send” but accidentally presses “Delete”
- C) User doesn’t know what an icon means, guesses wrong
- D) User misunderstands the whole system model

Answer: B

Explanation: Slip = right intention, wrong action (often inattention/skill). Mistake = wrong intention from wrong understanding.

MCQ 4

Which navigation goal is NOT one of the four golden rules?

- A) Knowing where you are
- B) Knowing what you can do

- C) Knowing how fast the app is
- D) Knowing where you've been

Answer: C

Explanation: The four are: where you are, what you can do, where you are going, where you've been.

MCQ 5

A "Hub and Spoke" navigation structure means:

- A) Users drill down through lists endlessly
- B) A central hub page is used to jump to other sections, returning to hub to switch
- C) One dataset is explored via filters
- D) All sections are visible on every page

Answer: B

Explanation: Hub-and-spoke uses a central hub as launchpad; to move between spokes, you go back to hub.

B) 5 Short Answer Questions (with answers)

SAQ 1

What does "design interactions, not just interfaces" mean? Give an example.

Answer:

It means design changes the *workflow and behavior*, not only the screen. Example: electronic stapling changes a repeated print-staple pattern into a "do work, print, then staple once" pattern.

SAQ 2

Explain "goals" and "constraints" in design with one example.

Answer:

Goals are the purpose and who it's for; constraints are limits like platform/time/budget. Example: Goal = make an app easy for warehouse staff; constraint = small phone screen + limited training time; trade-off = simplify features to keep it usable.

SAQ 3

Define persona and explain why details matter.

Answer:

A persona is a detailed description of an example user used as a surrogate during design. Details (age, background, fears, abilities) make them feel real so teams can design for real needs rather than guessing.

SAQ 4

What are scenarios used for, and what is the limitation of scenarios?

Answer:

Scenarios communicate design ideas and show step-by-step interaction flow. Limitation: they are linear, so one scenario can miss branches, choices, and error conditions; you need multiple scenarios.

SAQ 5

Explain STM and why “7±2 chunks” matters for interface design.

Answer:

Short-term/working memory is a temporary scratchpad with limited capacity (~7±2 chunks). Interfaces should not force users to remember too many items at once; grouping and structure reduce memory load.

C) 1 Case Study (with model answer)

Case Study:

You are redesigning a university “Course Registration” portal. Students complain they get lost, make mistakes, and panic when selecting courses.

Task: Using Week 6 concepts, propose improvements in:

1. navigation design
2. screen layout
3. reducing errors and cognitive load

Model Answer**1) Navigation design**

Apply the **four golden rules**:

- Show “where you are” using breadcrumbs like: Home > Registration > Select Courses.

- Clearly show “what you can do” with obvious buttons (“Add course”, “Drop course”, “View timetable”).
- Show “where you are going” with clear labels (“Continue to Confirmation”).
- Show “where you’ve been” using a progress indicator (Step 1/3, Step 2/3).

2) Screen layout

Use layout principles:

- Group related items: “Course search” separate from “Selected courses.”
- Order: top-to-bottom flow for data entry; tab order correct.
- Alignment: left-align text for scanning; align columns for course code/credits.
- White space: separate sections to reduce clutter and highlight important warnings.

3) Reducing errors and memory load

Design for predictable human error:

- Add confirmation for dangerous actions (“Drop course?”) to prevent slips.
- Provide undo for accidental drops.
- Reduce reliance on STM (7 ± 2 chunks) by showing chosen courses in a visible list instead of requiring memory.
- Use clear feedback to prevent wrong mental models (mistakes).